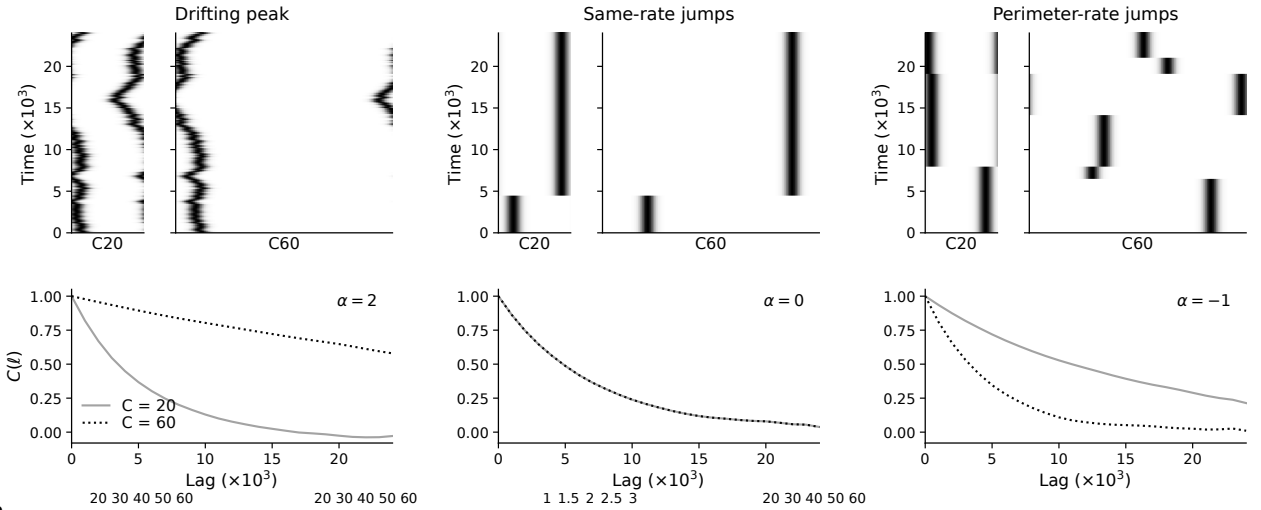
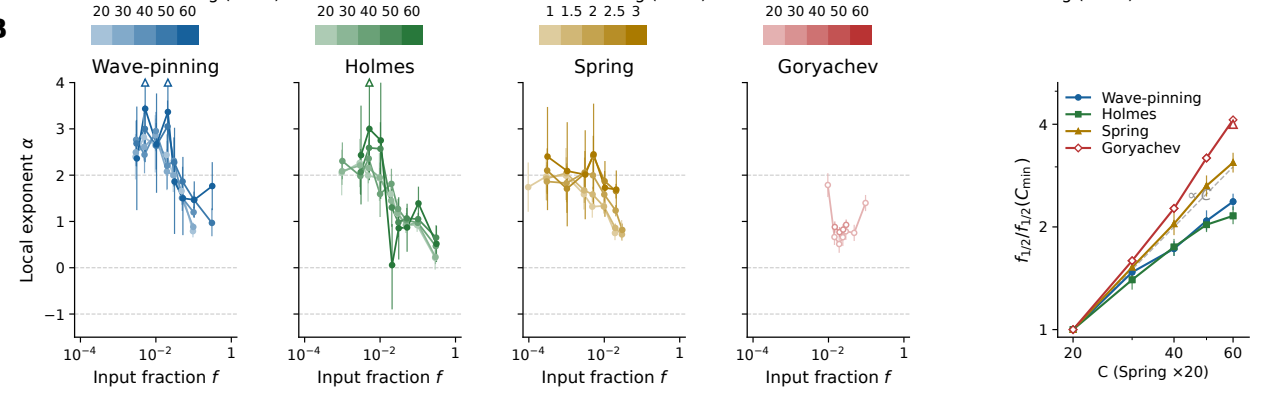
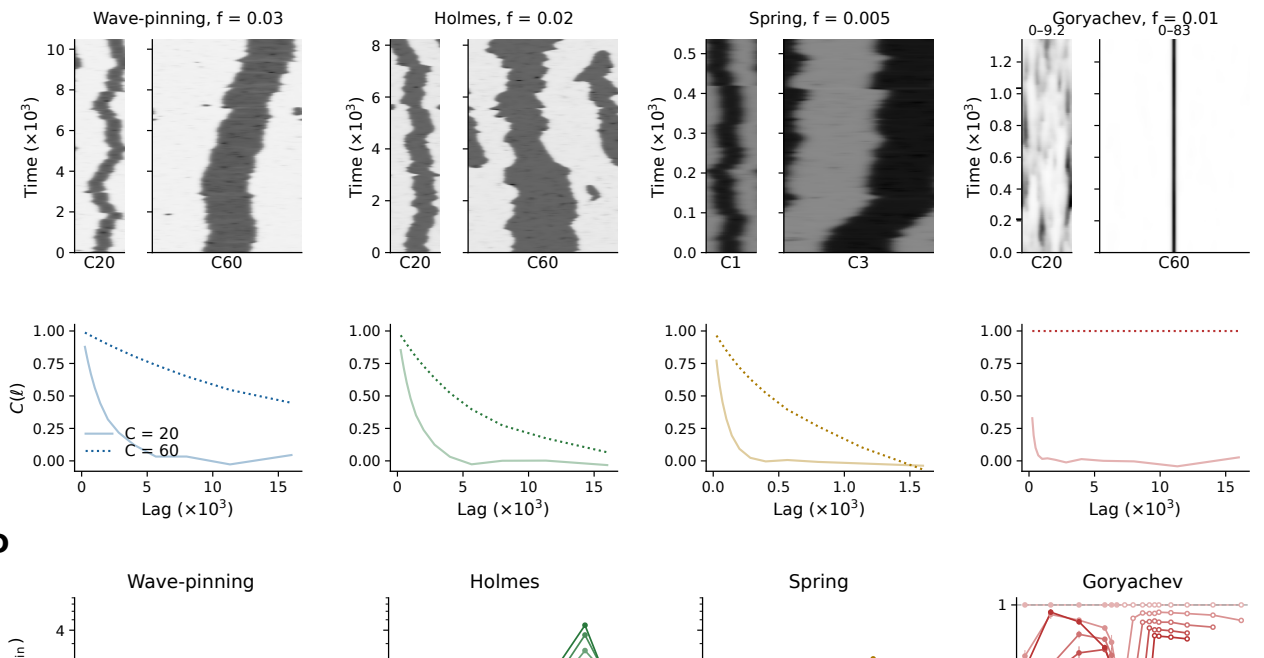
A


Drifting peak
Same-rate jumps
Perimeter-rate jumps
Time (×10³)
C20
C60
α = 2
α = 0
α = −1
C(ℓ)
C = 20
C = 60
Lag (×10³)

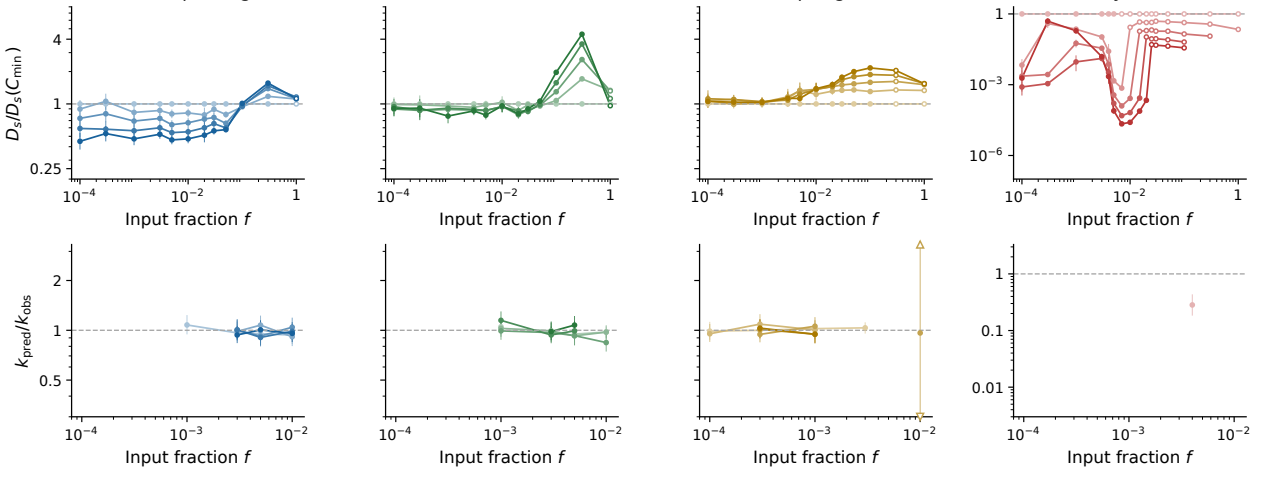
B


20 30 40 50 60
1 1.5 2 2.5 3
Wave-pinning
Holmes
Spring
Goryachev
Local exponent α
Input fraction f
Wave-pinning
Holmes
Spring
Goryachev
f1/2/f1/2(Cmin)
C (Spring ×20)

C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
0–9.2
0–83
Time (×10³)
C20
C60
C1
C3
C(ℓ)
C = 20
C = 60
Lag (×10³)

D


Wave-pinning
Holmes
Spring
Goryachev
Ds/Ds(Cmin)
kpred/kobs
Input fraction f

E

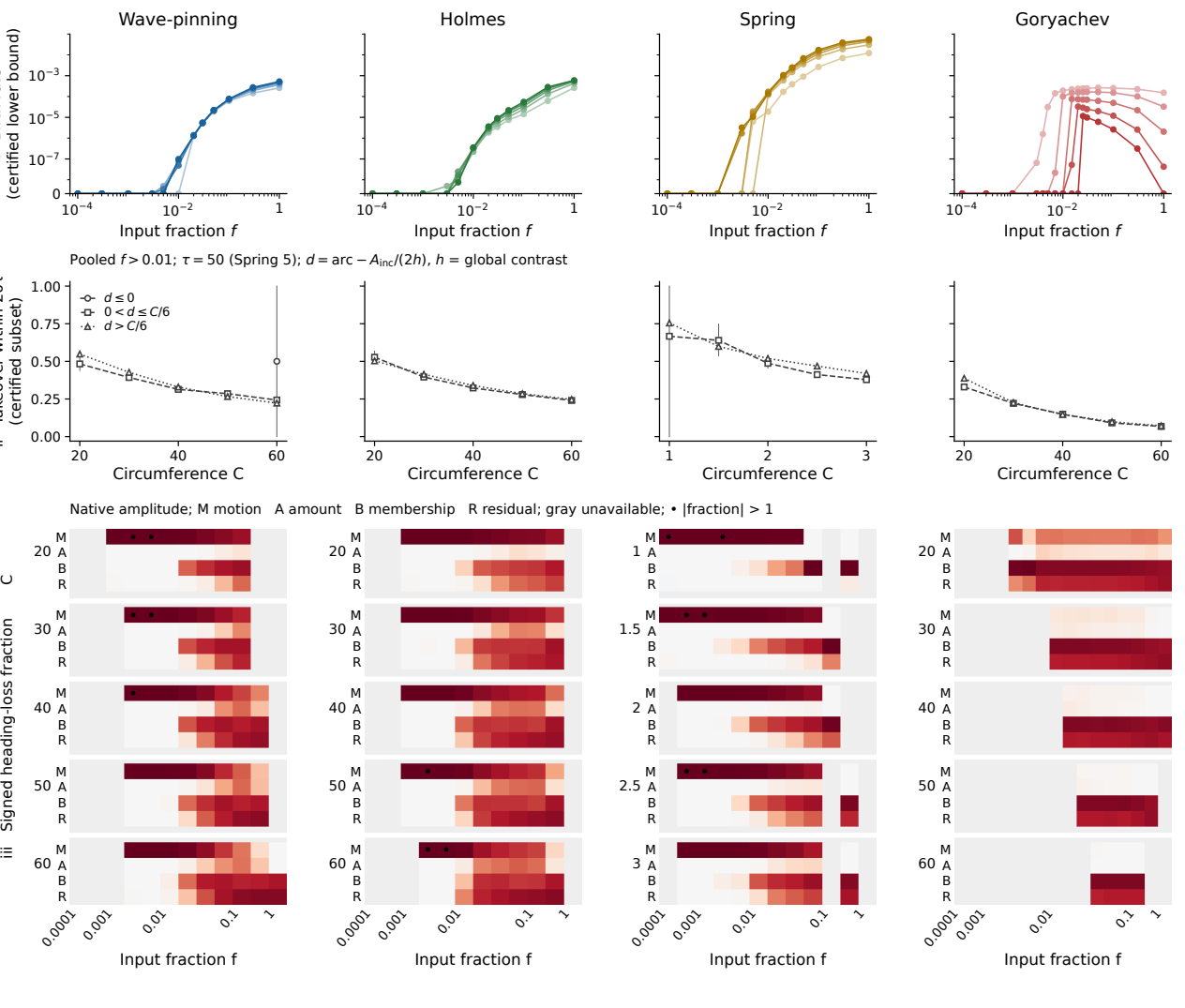

Wave-pinning
Holmes
Spring
Goryachev
i Birth rate (certified lower bound)
Input fraction f
Pooled f > 0.01; τ = 50 (Spring 5); d = arc − Ainc/(2h), h = global contrast
ii Takeover within 20τ (certified subset)
d ≤ 0
0 < d ≤ C/6
d > C/6
Circumference C
Native amplitude; M motion A amount B membership R residual; gray unavailable; • |fraction| > 1
iii Signed heading-loss fraction
C
M
A
B
R
Input fraction f

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics

A. Drifting peaks, same-rate null relocations and perimeter-rate null relocations give size exponents 2, 0 and −1. Null kymograph widths follow circumference; time increases upward. Autocorrelations C(ℓ) use solid C20 and dotted C60 curves. The null relocations are illustrative whole-peak moves, distinct from measured dominant-region replacements and birth-to-takeover events.

B. Larger rings retain heading across stronger fluctuating input. Input fraction f sets the ensemble mean to f times the native reference (Wave-pinning 0.1; Holmes 1; Spring 4; Goryachev 6.4). Input is positive lognormal, mean × $\exp(\sigma Z - \sigma^2/2)$, with unit-variance Gaussian driving field Z and σ=1. The spatial parameter ℓ and OU correlation time τ are 1.5 and 50 in chemical models; Spring uses basal-physical ℓ=0.15 and τ=5, with physical ℓ scaling with its changing circumference. Local size exponents α and noise-tolerance folds use model hues and light-to-dark circumference ramps. Open symbols in the exponent and drift panels identify conditions flagged as forced by the frozen matched-mean/lower-noise controls; open diamonds in the fold panel identify Goryachev irrespective of forcing; vertical intervals are pointwise 95% whole-history bootstrap intervals. Boundary triangles retain off-scale values. 125/260 local neighborhoods resolve, including 11 Goryachev neighborhoods. Largest-size tolerance folds are Wave-pinning 2.373, Holmes 2.155, Spring 3.088 and Goryachev 4.122. $f_{1/2}$ is the first input crossing of C(ℓ)=0.5 at reference lag 2000 (Spring 200); Spring circumference is multiplied by 20 only in the fold panel.

C. Matched small- and large-ring kymographs are centred on their initial activity heading to the nearest spatial bin. Time increases upward; black outside ticks mark persistent dominant-region replacements. Activity scales are shared within Wave-pinning, Holmes and Spring; Goryachev uses 0–9.2 at C20 and 0–83 at C60. Ensemble C(ℓ) curves are below. A low-contrast initial heading may have weak biological meaning despite a defined numerical centre.

D. Physical within-tenure drift D_s is normalized by the smallest size at the same input. Unity gives the equal-physical-drift geometric expectation α=2. Weak-input log–log fits for f≤0.01 give exponents Wave-pinning 1.96–1.99; Holmes 1.94–1.95; Spring 1.87–1.93. The second row shows the existing event-based predicted/observed decay only where f≤0.01 and multiple regions occupy at most 5% of frames. 45/47 eligible point comparisons lie within 15% (Wave-pinning 16/16; Holmes 15/16; Spring 14/14; Goryachev 0/1). These point predictions assign the scored-event mean angular loss to unscored replacements; this remains an assumption. Goryachev uses separate logarithmic ordinate ranges: 10^{-7}–2 for normalized drift and 0.003–3.3 for decay ratio; other models use 0.2–8 and 0.3–3.3. The actual mean circumference, $\bar{C}$, enters physical-to-angular conversion. Figure 3D independently predicts this measured drift from deterministic edge responses.

E. i, Certified no-parent secondary births per actual perimeter-time: new tracks persist at least five frames, have a preceding incumbent, and have conservative support enclosures disjoint from all previous regions. Rates are lower bounds; discarded masks prevent exact birth/split classification. ii, Observed incidence of five-frame global dominance by 20τ (1000 chemical units; 100 Spring), conditional on certified births with that follow-up. Inputs f>0.01 are pooled.

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics · continued

Distance d subtracts estimated incumbent half-width (preceding full-basin amount divided by twice global contrast h) from the shortest centroid arc. Vertical intervals are 95% whole-history bootstrap intervals. A lower finite-horizon incidence can reflect delayed or less frequent takeover; lifetime success remains unidentified. Figure 2B–C motivates separating initiation from incumbent removal. iii, Native-amplitude-aware signed heading-loss allocation over the condition’s analysis lag. M, shared-region motion; A, coexisting amount; B, track membership including unresolved splits/mergers; R, amplitude/shape/background remainder. Point-region Shapley changes are scaled by mean native endpoint moment amplitude, so pure rigid translation gives M=1, R=0. Components sum exactly to observed heading loss but are descriptive, convention-dependent shares. The symmetric-log color scale is linear within ±0.05 and spans ±1; dots indicate values outside ±1, gray unavailable pairs. The unit-normalized point-region diagnostic and its normalization artefact are preserved in the supplement; eligibility and equations are detailed in Methods.

$$Q_t = \frac{\sum_{i \in I_t} A_{i,t} e^{i\theta_{i,t}}}{\sum_{i \in I_t} A_{i,t}}, \qquad r_t = |P_t|, \quad \widetilde{Q}_t = r_t Q_t \tag{4.1}$$

Q is the normalized point-region moment; P=M₁/M₀ is measured from the native field. Its amplitude r prevents the unit-modulus single-region normalization artefact.

$$\Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}), \qquad \Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q \tag{4.2}$$

The three ΔQ components use the symmetric Shapley convention for motion, coexisting amount and membership; r̄ is mean endpoint native amplitude. R closes the native moment change and includes amplitude variation, shape and background. This is an accounting convention, not causal attribution.

$$L_j = \frac{\mathrm{Re}\left[\Delta P_j \, \overline{(H_1 - H_0)}\right]}{|P_0| + |P_1|}, \quad H_t = \frac{P_t}{|P_t|}, \quad \sum_j L_j = 1 - \cos(\theta_1 - \theta_0) \tag{4.3}$$

Pooled signed fractions are sums of L_j divided by total measured heading loss. They retain cancellation, can exceed one, and depend on the stated accounting convention; they are not causal shares.

$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.4}$$

C(ℓ) is activity-heading autocorrelation; C̄ is actual mean circumference. Endpoint eligibility uses |P|≥0.1. Drift closure is shown only at f≤0.01 and multiple-region fraction≤0.05.